```
EditURL = "../../examples/demo.jl"
```

HARE.jl – Tutorial

This tutorial demonstrates estimators in HARE.jl using the formula interface. Every function also accepts a matrix interface `f(X, y; ...)` where `X` is the regressor matrix **without** a constant column; both signatures are shown for the first estimator and listed for all estimators in the API Reference. Each section simulates data appropriate for that estimator.

Setup

$$y_i = \mathbf{x}_i^\top \beta + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0,\, \sigma_i^2)$$

For this tutorial, as exogenous variables we generate two sets of random samples ($\mathbf{x}_1$) and ($\mathbf{x}_2$), normally distributed. The response variable (y) is given by

$$y_i = \beta_0 + \beta_1\, \mathbf{x}_{1i} + \beta_2\, \mathbf{x}_{2i} + \varepsilon_i$$

```
using HARE
using Random, LinearAlgebra, GLM, StatsBase

Random.seed!(1234)

nobs   = 500
x1     = randn(nobs)
x2     = randn(nobs)
b = [1.0, 2.0, -1.0];              # beta true parameters
```

Exponential Variance Model:

The Harvey (1976) model specifies multiplicative heteroskedasticity via an exponential variance function:

$$\sigma_i^2 = \sigma_0^2 \exp(\mathbf{z}_i^\top \gamma)$$

We simulate data with log variance proportional to $|\mathbf{x}_1|$ and $\sigma_0 = 1$.

```
sigma_0 = 1.0;
gamma   = [1.0, 0.0];
sigma_exp = sigma_0 * exp.((gamma[1] .* abs.(x1) .+ gamma[2] .* x2) ./ 2);
```

The simulated dataset ($data$) is

```
y    = b[1] .+ b[2] .* x1 .+ b[3] .* x2 .+ sigma_exp .* randn(nobs);
data = (y = y, x1 = x1, x2 = x2);
```

MLE (`exponential_mle`): jointly maximises the log-likelihood over (β, γ) via L-BFGS, warm-started from the Harvey auxiliary regression. Equivalent to R's `nlme::gls(weights = varComb(varExp(...)))`.

```
exponential_mle(formula, data; Z=nothing) -> MLEResult
```

```
exponential_mle(@formula(y ~ x1 + x2), data; Z = @formula(0 ~ abs(x1) + x2))
```

```
Coefficients:
```

	Coef.	Std. Error	t	Pr(>|t|)	Lower 95%	Upper 95%
(Intercept)	1.00997	0.0632213	15.98	<1e-45	0.885755	1.13418

```
x1            1.90611   0.0881646   21.62    <1e-72   1.73289    2.07933
x2           -1.02156   0.061346   -16.65    <1e-49  -1.14209   -0.901027
──────────────────────────────────────────────────────────────────────────

Link: Exponential — σ²ᵢ = exp(γ₀ + zᵢ′γ)    [Harvey 1976]
Log-variance equation (gamma):
──────────────────────────────────────────────────────────────────────────
              Coef.  Std. Error      z  Pr(>|z|)  Lower 95%  Upper 95%
──────────────────────────────────────────────────────────────────────────
log σ₀²  -0.0344842   0.106269   -0.32    0.7456  -0.242768  0.1738
z1        1.05254     0.0988821  10.64    <1e-25   0.85873   1.24634
z2       -0.095787    0.0637538  -1.50    0.1330  -0.220742  0.0291681
──────────────────────────────────────────────────────────────────────────

Log-likelihood: -927.5787   Converged: true   Iterations: 10
N: 500   R^2: 0.617   Adj. R^2: 0.6154
```

Two-step FWLS (`exponential_fwls`): a closed-form alternative that requires no numerical optimisation. Preferable when MLE fails to converge or when the normality assumption is in doubt, since FWLS is consistent under any error distribution with the correct variance model.

```
exponential_fwls(formula, data; Z=nothing) -> FWLSResult
```

```
exponential_fwls(@formula(y ~ x1 + x2), data; Z = @formula(0 ~ abs(x1) + x2))
```

```
Coefficients:
──────────────────────────────────────────────────────────────────────────
                Coef.  Std. Error       t  Pr(>|t|)  Lower 95%  Upper 95%
──────────────────────────────────────────────────────────────────────────
(Intercept)   1.01124   0.0837127   12.08    <1e-28   0.846768   1.17572
x1            1.90355   0.12354     15.41    <1e-43   1.66082    2.14627
x2           -1.02852   0.08088    -12.72    <1e-31  -1.18743   -0.869612
──────────────────────────────────────────────────────────────────────────

Log-variance equation (gamma):
──────────────────────────────────────────────────────────────────────────
              Coef.  Std. Error      z  Pr(>|z|)  Lower 95%  Upper 95%
──────────────────────────────────────────────────────────────────────────
log σ₀²  -0.196615    0.169205  -1.16    0.2452  -0.528251  0.135021
z1        1.21344     0.15656    7.75    <1e-14   0.90659   1.5203
z2       -0.100246    0.101157  -0.99    0.3217  -0.29851   0.0980186
──────────────────────────────────────────────────────────────────────────

N: 500   R^2: 0.6169   Adj. R^2: 0.6154
```

Matrix interface (available for every estimator; `X` and `Z` without a constant column):

```
X   = hcat(x1, x2);
Z_h = hcat(abs.(x1), x2);
exponential_fwls(X, y; Z = Z_h)
```

```
Coefficients:
──────────────────────────────────────────────────────────────────────────
                Coef.  Std. Error       t  Pr(>|t|)  Lower 95%  Upper 95%
──────────────────────────────────────────────────────────────────────────
(Intercept)   1.01124   0.0837127   12.08    <1e-28   0.846768   1.17572
x1            1.90355   0.12354     15.41    <1e-43   1.66082    2.14627
x2           -1.02852   0.08088    -12.72    <1e-31  -1.18743   -0.869612
──────────────────────────────────────────────────────────────────────────

Log-variance equation (gamma):
──────────────────────────────────────────────────────────────────────────
```

	Coef.	Std. Error	z	Pr(>|z|)	Lower 95%	Upper 95%
log σ₀²	-0.196615	0.169205	-1.16	0.2452	-0.528251	0.135021
z1	1.21344	0.15656	7.75	<1e-14	0.90659	1.5203
z2	-0.100246	0.101157	-0.99	0.3217	-0.29851	0.0980186

N: 500 R^2: 0.6169 Adj. R^2: 0.6154

Power Variance Model

The power variance model (Harvey 1976, multiplicative form) specifies:

$$\sigma_i^2 = \sigma^2 \prod_j |z_{ij}|^{2\delta_j}$$

Equivalently, $\log \sigma_i^2 = \log \sigma^2 + 2\delta_1 \log |z_{i1}| + \cdots$, which is the Harvey exponential model applied to $\log |Z|$. The reported $\gamma = (\sigma^2, \delta_1, \ldots)$ is obtained via the delta-method reparametrisation.

We simulate data with $\sigma_i^2 = |x_1|$ (i.e. $\sigma^2 = 1$, $\delta_1 = 0.5$).

```
sigma_pow = abs.(x1) .^ 0.5;
y_pow     = b[1] .+ b[2] .* x1 .+ b[3] .* x2 .+ sigma_pow .* randn(nobs);
data_pow  = (y = y_pow, x1 = x1, x2 = x2);
```

MLE (`power_mle`): joint MLE warm-started from the Harvey auxiliary regression on $\log |Z|$. Reports $(\sigma^2, \delta_1, \ldots)$ with delta-method standard errors.

```
power_mle(@formula(y ~ x1 + x2), data_pow; Z = @formula(0 ~ abs(x1)))
```

Coefficients:

	Coef.	Std. Error	t	Pr(>|t|)	Lower 95%	Upper 95%
(Intercept)	1.00317	0.0195705	51.26	<1e-99	0.964722	1.04162
x1	1.98875	0.0474852	41.88	<1e-99	1.89546	2.08205
x2	-0.999924	0.017665	-56.60	<1e-99	-1.03463	-0.965217

Link: Power — log σ²ᵢ = γ₀ + Σγⱼlog\|zᵢⱼ\|
Power variance equation (log σ², 2δ):

	Coef.	Std. Error	z	Pr(>|z|)	Lower 95%	Upper 95%
log σ²	-0.0306619	0.0709218	-0.43	0.6655	-0.169666	0.108342
z1	1.01287	0.0564796	17.93	<1e-71	0.902172	1.12357

Log-likelihood: -557.9243 Converged: true Iterations: 12
N: 500 R^2: 0.8743 Adj. R^2: 0.8738

Two-step FWLS (`power_fwls`):

```
power_fwls(@formula(y ~ x1 + x2), data_pow; Z = @formula(0 ~ abs(x1)))
```

Coefficients:

	Coef.	Std. Error	t	Pr(>|t|)	Lower 95%	Upper 95%
(Intercept)	1.00132	0.011958	83.74	<1e-99	0.977826	1.02481
x1	1.99449	0.022658	88.03	<1e-99	1.94997	2.03901

```
x2           -1.00398   0.0109698  -91.52    <1e-99  -1.02553   -0.982426
```

```
Power variance equation (log σ², 2δ):
```

	Coef.	Std. Error	z	Pr(>|z|)	Lower 95%	Upper 95%
log σ²	-0.210162	0.117995	-1.78	0.0749	-0.441428	0.0211047
z1	0.841352	0.093567	8.99	<1e-18	0.657964	1.02474

```
N: 500   R^2: 0.8744   Adj. R^2: 0.8739
```

Quadratic Variance Model

The Glejser (1969) model assumes the standard deviation is linear in auxiliary regressors $\mathbf{z}_i$:

$$\sigma_i = \sigma_0 + \mathbf{z}_i^\top \gamma$$

We simulate data with σ_i linear in $|\mathbf{x}_1|$.

Because the variance depends on $|\mathbf{x}_1|$, not $\mathbf{x}_1$ itself, we supply a custom auxiliary regressor matrix `Z = [|x1|, x2]` so the Glejser auxiliary regression is correctly specified. Using the raw regressors would yield a badly biased gamma.

```
sigma = 0.5 .+ 0.3 .* abs.(x1);
y = b[1] .+ b[2] .* x1 .+ b[3] .* x2 .+ sigma .* randn(nobs);
data = (y = y, x1 = x1, x2 = x2);
quadratic_mle(@formula(y ~ x1 + x2), data; Z = @formula(0 ~ abs(x1) + x2))
```

```
Coefficients:
```

	Coef.	Std. Error	t	Pr(>|t|)	Lower 95%	Upper 95%
(Intercept)	1.01448	0.0301162	33.69	<1e-99	0.955305	1.07365
x1	2.03922	0.0399564	51.04	<1e-99	1.96072	2.11772
x2	-1.04103	0.02877	-36.18	<1e-99	-1.09755	-0.984501

```
Link: Quadratic   − σᵢ  = γ₀ + zᵢ′γ          [Glejser 1969]
Std. deviation equation (gamma):
```

	Coef.	Std. Error	z	Pr(>|z|)	Lower 95%	Upper 95%
σ₀	0.456078	0.0292284	15.60	<1e-54	0.398792	0.513365
z1	0.343226	0.0380666	9.02	<1e-18	0.268617	0.417835
z2	0.0137392	0.021326	0.64	0.5194	-0.0280591	0.0555374

```
Log-likelihood: -546.9842   Converged: true   Iterations: 11
N: 500   R^2: 0.9003   Adj. R^2: 0.8999
```

Linear Variance Model

The linear variance model specifies the variance as a linear combination of auxiliary regressors $\mathbf{z}_i$:

$$\sigma_i^2 = \gamma_0 + \mathbf{z}_i^\top \gamma$$

We simulate data with σ_i^2 linear in $|\mathbf{x}_1|$.

```
sigma2 = 0.5 .+ 0.3 .* abs.(x1);
y = b[1] .+ b[2] .* x1 .+ b[3] .* x2 .+ sqrt.(sigma2) .* randn(nobs);
data = (y = y, x1 = x1, x2 = x2);
linear_mle(@formula(y ~ x1 + x2), data; Z = @formula(0 ~ abs(x1) + x2))
```

```
Coefficients:
```

	Coef.	Std. Error	t	Pr(>|t|)	Lower 95%	Upper 95%
(Intercept)	1.06012	0.0388198	27.31	<1e-99	0.983849	1.13639
x1	2.05141	0.0409823	50.06	<1e-99	1.97089	2.13193
x2	-1.00749	0.0375162	-26.85	<1e-98	-1.0812	-0.933782

```
Link: Linear      — σ²ᵢ = γ₀ + zᵢ′γ
Variance equation (gamma):
```

	Coef.	Std. Error	z	Pr(>|z|)	Lower 95%	Upper 95%
σ₀²	0.55191	0.0783311	7.05	<1e-11	0.398384	0.705436
z1	0.266642	0.0968623	2.75	0.0059	0.0767957	0.456489
z2	0.0289895	0.0432038	0.67	0.5022	-0.0556885	0.113667

```
Log-likelihood: -642.3411   Converged: true   Iterations: 4
N: 500   R^2: 0.8856   Adj. R^2: 0.8851
```

Groupwise Heteroskedasticity

The groupwise model assumes a constant variance within each group g:

$$\sigma_i^2 = \sigma_g^2, \quad i \in g$$

We simulate four groups with increasing variance.

```
groups   = repeat(["A", "B", "C", "D"], inner = nobs ÷ 4);
sigma_gw = [0.5, 1.0, 1.5, 2.0];
sigma_g  = [sigma_gw[findfirst(==( g), ["A","B","C","D"])] for g in groups];
y_gw     = b[1] .+ b[2] .* x1 .+ b[3] .* x2 .+ sigma_g .* randn(nobs);
data_gw  = (y = y_gw, x1 = x1, x2 = x2);
```

MLE (`groupwise_mle`): exact MLE standard errors from $(X^\top W X)^{-1}$:

```
groupwise_mle(@formula(y ~ x1 + x2), data_gw, groups)
```

```
Coefficients:
```

	Coef.	Std. Error	t	Pr(>|t|)	Lower 95%	Upper 95%
(Intercept)	0.997978	0.0354523	28.15	<1e-99	0.928323	1.06763
x1	2.0166	0.0326343	61.79	<1e-99	1.95249	2.08072
x2	-0.944209	0.0342432	-27.57	<1e-99	-1.01149	-0.876929

```
Group variances (G = 4):
```

Group	n_g	σ²	σ
A	125	0.218385	0.467317
B	125	0.90287	0.950195

```
  C                   125      2.254301    1.501433
  D                   125      4.065849    2.016395

Log-likelihood: -746.4561   Converged: true   Iterations: 6
N: 500   R^2: 0.7613   Adj. R^2: 0.7603
```

Iterated FWLS (`iterated_groupwise`): identical point estimates, GLM-scaled standard errors:

```
iterated_groupwise(@formula(y ~ x1 + x2), data_gw, groups)
```

```
Coefficients:
──────────────────────────────────────────────────────────────────────────────
                 Coef.  Std. Error       t  Pr(>|t|)  Lower 95%  Upper 95%
──────────────────────────────────────────────────────────────────────────────
(Intercept)   0.997978   0.0281327   35.47    <1e-99   0.942704    1.05325
x1            2.0166     0.0258966   77.87    <1e-99   1.96572     2.06748
x2           -0.944209   0.0271733  -34.75    <1e-99  -0.997597   -0.89082
──────────────────────────────────────────────────────────────────────────────
Group variances (G = 4):
  Group              n_g            σ²           σ
  ──────────────────────────────────────────────────
  A                  125      0.218385    0.467317
  B                  125       0.90287    0.950195
  C                  125      2.254301    1.501433
  D                  125      4.065849    2.016395

Log-likelihood: -746.4561   Converged: true   Iterations: 6
N: 500   R^2: 0.7613   Adj. R^2: 0.7603
```

AR(1) Estimators

We simulate data with AR(1) errors: $u_t = 0.7\,u_{t-1} + e_t$.

```
eps  = randn(nobs);
u_ar = zeros(nobs);
for t in 2:nobs; u_ar[t] = 0.7 * u_ar[t-1] + eps[t]; end
y_ar    = b[1] .+ b[2] .* x1 .+ b[3] .* x2 .+ u_ar;
data_ar = (y = y_ar, x1 = x1, x2 = x2);
```

Prais-Winsten – retains the first observation via GLS transformation:

```
two_step_prais_winsten(@formula(y ~ x1 + x2), data_ar)
```

```
Coefficients:
──────────────────────────────────────────────────────────────────────────────
                 Coef.  Std. Error       t  Pr(>|t|)  Lower 95%  Upper 95%
──────────────────────────────────────────────────────────────────────────────
(Intercept)   1.04781   0.156339     6.70    <1e-10   0.740645    1.35498
x1            1.98397   0.0321344   61.74    <1e-99   1.92083     2.04711
x2           -0.97822   0.0344398  -28.40    <1e-99  -1.04589    -0.910554
──────────────────────────────────────────────────────────────────────────────
AR(1) coefficient:
──────────────────────────────────────────────────────────────────────
      Coef.  Std. Error      z  Pr(>|z|)  Lower 95%  Upper 95%
──────────────────────────────────────────────────────────────────────
ρ  0.727404   0.0306882  23.70    <1e-99   0.667256   0.787552
──────────────────────────────────────────────────────────────────────
N: 500   R^2: 0.7477   Adj. R^2: 0.7467
```

Hildreth-Lu – grid search over $\rho \in (-0.99, 0.99)$:

```
hildreth_lu(@formula(y ~ x1 + x2), data_ar)
```

```
Coefficients:
```

	Coef.	Std. Error	t	Pr(>|t|)	Lower 95%	Upper 95%
(Intercept)	1.04824	0.159783	6.56	<1e-09	0.73431	1.36218
x1	1.98394	0.0321171	61.77	<1e-99	1.92084	2.04705
x2	-0.978189	0.0344114	-28.43	<1e-99	-1.0458	-0.910579

```
AR(1) coefficient:
```

	Coef.	Std. Error	z	Pr(>|z|)	Lower 95%	Upper 95%
ρ	0.731307	0.0305022	23.98	<1e-99	0.671523	0.79109

```
Grid points: 200
N: 500   R^2: 0.7477   Adj. R^2: 0.7467
```

Beach-MacKinnon – exact maximum likelihood:

```
beach_mackinnon(@formula(y ~ x1 + x2), data_ar)
```

```
Coefficients:
```

	Coef.	Std. Error	t	Pr(>|t|)	Lower 95%	Upper 95%
(Intercept)	1.04781	0.155994	6.72	<1e-10	0.741318	1.35429
x1	1.98398	0.0321437	61.72	<1e-99	1.92082	2.04713
x2	-0.978225	0.0344497	-28.40	<1e-99	-1.04591	-0.91054

```
AR(1) coefficient:
```

	Coef.	Std. Error	z	Pr(>|z|)	Lower 95%	Upper 95%
ρ	0.726796	0.030717	23.66	<1e-99	0.666591	0.787

```
Log-likelihood: 22.6585   Converged: true
N: 500   R^2: 0.7477   Adj. R^2: 0.7467
```

R Cross-Validation

Each MLE estimator has an exact R counterpart in `nlme::gls`. Below we run both on the same dataset — generated once in Julia with `StableRNG` and saved to CSV so R can read it back — so the output tables can be compared directly. The data uses `StableRNG(42)` for the predictors and `StableRNG(99)` for the noise.

```
using StableRNGs
rng_r  = StableRNG(42);
rng_r2 = StableRNG(99);
n_r    = 500;
x1_r   = randn(rng_r, n_r);
x2_r   = randn(rng_r, n_r);
Xr     = hcat(x1_r, x2_r);
br     = [1.0, 2.0, -1.0];
```

Exponential MLE

R code:

```
library(nlme)
d <- read.csv("hare_exp.csv")   # x1, x2, y — written from Julia
m <- gls(y ~ x1 + x2, data = d, weights = varExp(form = ~x1), method = "ML")
summary(m)
```

R output:

```
Generalized least squares fit by maximum likelihood
  Model: y ~ x1 + x2
  Data: d
       AIC      BIC    logLik
  1399.339 1420.412 -694.6695

Variance function:
 Structure: Exponential of variance covariate
 Formula: ~x1
 Parameter estimates:
    expon
0.5400336

Coefficients:
                 Value Std.Error   t-value p-value
(Intercept)  0.9762948 0.0494178  19.75594       0
x1           1.9933986 0.0311266  64.04173       0
x2          -1.0114907 0.0298082 -33.93332       0

Residual standard error: 1.001954
Degrees of freedom: 500 total; 497 residual
```

HARE: `gamma[1]` $= \log\sigma^2 =$ `log(sigma^2)` from R; `gamma[2]` $= 2\delta =$ `2 * 0.5400336`:

```
y_r_exp = br[1] .+ br[2] .* x1_r .+ br[3] .* x2_r .+ exp.(0.5 .* x1_r) .* randn(rng_r2, n_r);
exponential_mle(Xr, y_r_exp; Z = hcat(x1_r))
```

```
Coefficients:
──────────────────────────────────────────────────────────────────────────────
                 Coef.  Std. Error       t  Pr(>|t|)  Lower 95%  Upper 95%
──────────────────────────────────────────────────────────────────────────────
(Intercept)   0.976295   0.0492883   19.81    <1e-64   0.879456   1.07313
x1            1.9934     0.0310571   64.19    <1e-99   1.93238    2.05442
x2           -1.01149    0.029809   -33.93    <1e-99  -1.07006   -0.952924
──────────────────────────────────────────────────────────────────────────────
Link: Exponential — σ²ᵢ = exp(γ₀ + zᵢ′γ)    [Harvey 1976]
Log-variance equation (gamma):
──────────────────────────────────────────────────────────────────────────
              Coef.  Std. Error      z  Pr(>|z|)  Lower 95%  Upper 95%
──────────────────────────────────────────────────────────────────────────
log σ₀²  0.00390501   0.0633408   0.06    0.9508  -0.120241   0.128051
z1       1.08007      0.0594226  18.18    <1e-73   0.963601   1.19653
──────────────────────────────────────────────────────────────────────────
Log-likelihood: -694.6695   Converged: true   Iterations: 10
N: 500   R^2: 0.738   Adj. R^2: 0.7369
```

Power MLE

R code:

```
d <- read.csv("hare_pow.csv")
m <- gls(y ~ x1 + x2, data = d, weights = varPower(form = ~abs(x1)), method = "ML")
summary(m)
```

R output:

```
Generalized least squares fit by maximum likelihood
  Model: y ~ x1 + x2
  Data: d
       AIC      BIC    logLik
  1141.608 1162.681 -565.8042

Variance function:
 Structure: Power of variance covariate
 Formula: ~abs(x1)
 Parameter estimates:
    power
0.4773703

Coefficients:
                 Value Std.Error   t-value p-value
(Intercept)  0.9955270 0.0228240  43.61754       0
x1           2.1150261 0.0483900  43.70791       0
x2          -0.9639498 0.0242141 -39.80943       0

Residual standard error: 0.98068
Degrees of freedom: 500 total; 497 residual
```

HARE: `gamma[1]` $= \log \sigma^2$; `gamma[2]` $= 2\delta =$ `2 * 0.4773703`:

```
y_r_pow = br[1] .+ br[2] .* x1_r .+ br[3] .* x2_r .+ abs.(x1_r).^0.5 .* randn(rng_r2, n_r);
power_mle(Xr, y_r_pow; Z = hcat(abs.(x1_r)))
```

```
Coefficients:
──────────────────────────────────────────────────────────────────────────────
                 Coef.  Std. Error       t  Pr(>|t|)  Lower 95%  Upper 95%
──────────────────────────────────────────────────────────────────────────────
(Intercept)   0.995527   0.0227699   43.72    <1e-99    0.95079   1.04026
x1            2.11503    0.0483153   43.78    <1e-99    2.0201    2.20995
x2           -0.96395    0.0241777  -39.87    <1e-99   -1.01145  -0.916447
──────────────────────────────────────────────────────────────────────────────
Link: Power       − log σ²ᵢ = γ₀ + Σγⱼlog|zᵢⱼ|
Power variance equation (log σ², 2δ):
──────────────────────────────────────────────────────────────────────────────
             Coef.  Std. Error      z  Pr(>|z|)  Lower 95%  Upper 95%
──────────────────────────────────────────────────────────────────────────────
log σ²  -0.0390182   0.0750069  -0.52    0.6029  -0.186029   0.107993
z1       0.954741    0.0718746  13.28    <1e-39   0.813869   1.09561
──────────────────────────────────────────────────────────────────────────────
Log-likelihood: -565.8042   Converged: true   Iterations: 11
N: 500   R^2: 0.8838   Adj. R^2: 0.8833
```

Groupwise MLE

R code:

```
d <- read.csv("hare_gw.csv")
d$group <- factor(d$group)
m <- gls(y ~ x1 + x2, data = d, weights = varIdent(form = ~1 | group), method = "ML")
summary(m)
```

R output:

```
Generalized least squares fit by maximum likelihood
  Model: y ~ x1 + x2
  Data: d
       AIC      BIC    logLik
  1464.294 1493.797 -725.1472

Variance function:
 Structure: Different standard deviations per stratum
 Formula: ~1 | group
 Parameter estimates:
       1        2        3        4
1.000000 1.661873 2.666309 3.399754

Coefficients:
                 Value Std.Error   t-value p-value
(Intercept)  0.9999977 0.0373464  26.77628       0
x1           1.9740476 0.0362961  54.38735       0
x2          -0.9412746 0.0376907 -24.97368       0

Residual standard error: 0.5237556
Degrees of freedom: 500 total; 497 residual
```

R's `varIdent` reports relative SDs per group (group 1 = reference = 1.0). HARE reports σ_g^2 directly for each group:

```
gw_r_groups = repeat(["A","B","C","D"], inner = n_r ÷ 4);
gw_r_sig    = Dict("A"=>0.5,"B"=>1.0,"C"=>1.5,"D"=>2.0);
y_r_gw      = br[1] .+ br[2] .* x1_r .+ br[3] .* x2_r .+ [gw_r_sig[g] for g in gw_r_groups] .* randn(rng_r2, n_r);
groupwise_mle(Xr, y_r_gw, gw_r_groups)
```

```
Coefficients:
──────────────────────────────────────────────────────────────────────────────
                Coef.  Std. Error       t  Pr(>|t|)  Lower 95%  Upper 95%
──────────────────────────────────────────────────────────────────────────────
(Intercept)   0.999998   0.0372344   26.86    <1e-98   0.926842   1.07315
x1             1.97405   0.0361872   54.55    <1e-99   1.90295    2.04515
x2           -0.941275   0.0375776  -25.05    <1e-89  -1.01511   -0.867444
──────────────────────────────────────────────────────────────────────────────

Group variances (G = 4):
  Group              n_g            σ²           σ
  ──────────────────────────────────────────────────
  A                  125      0.274323    0.523759
  B                  125      0.757638    0.870424
  C                  125      1.950139    1.396474
  D                  125       3.17067    1.780638
```

```
Log-likelihood: -725.1472   Converged: true   Iterations: 5
N: 500   R^2: 0.7556   Adj. R^2: 0.7547
```

Consistency: Exponential MLE

With a large sample the MLE should recover the true parameters $\beta = [1, 2, -1]$ and $\gamma = [0, 1, 0]$ (intercept $\log \sigma_0^2 = 0$, slope on x_1 equal to 1, slope on x_2 equal to 0).

```
rng_c   = StableRNG(42);
N       = 1_000_000;
x1_c    = randn(rng_c, N);
x2_c    = randn(rng_c, N);
sigma_c = exp.(0.5 .* x1_c);
y_c     = 1.0 .+ 2.0 .* x1_c .- 1.0 .* x2_c .+ sigma_c .* randn(rng_c, N);
data_c  = (y = y_c, x1 = x1_c, x2 = x2_c);

m_c = exponential_mle(@formula(y ~ x1 + x2), data_c)
```

```
Coefficients:
```

	Coef.	Std. Error	t	Pr(>|t|)	Lower 95%	Upper 95%
(Intercept)	1.00085	0.00109866	910.97	<1e-99	0.998701	1.00301
x1	2.00132	0.00077413	2585.26	<1e-99	1.99981	2.00284
x2	-0.999302	0.000777795	-1284.79	<1e-99	-1.00083	-0.997778

```
Link: Exponential — σ²ᵢ = exp(γ₀ + zᵢ′γ)    [Harvey 1976]
Log-variance equation (gamma):
```

	Coef.	Std. Error	z	Pr(>|z|)	Lower 95%	Upper 95%
log σ₀²	-0.00301577	0.00141422	-2.13	0.0330	-0.00578758	-0.000243958
z1	0.998305	0.0014144	705.82	<1e-99	0.995533	1.00108
z2	7.49778e-5	0.00141193	0.05	0.9576	-0.00269235	0.0028423

```
Log-likelihood: -1.4174502441e6   Converged: false   Iterations: 13
N: 1000000   R^2: 0.7534   Adj. R^2: 0.7534
```

Combined AR(1) + Heteroskedasticity

The DGP is $u_t = \rho\, u_{t-1} + \sigma_t\, \varepsilon_t$ with Harvey-type heteroskedasticity in the innovations. True parameters: $\rho = 0.7$, $\gamma = [1,\ 0]$ on $(|x_1|, x_2)$ (reusing `sigma_exp` from the Exponential Variance section above).

Both estimators run the log-variance regression in the original space: the Sequential estimator regresses the log squared Prais-Winsten residuals on $\mathbf{Z}$, since the innovations $\sigma_t \varepsilon_t$ have variance $\sigma_0^2 \exp(\mathbf{z}_t^\top \gamma)$; the Joint estimator maximises the exact likelihood. As in the sections above, `Z = [|x1|, x2]` is passed, so both report γ for the variables the variance depends on.

```
u_hae = zeros(nobs);
for t in 2:nobs; u_hae[t] = 0.7 * u_hae[t-1] + sigma_exp[t] * eps[t]; end
y_hae    = b[1] .+ b[2] .* x1 .+ b[3] .* x2 .+ u_hae;
data_hae = (y = y_hae, x1 = x1, x2 = x2);
```

Sequential HARE – corrects for AR(1) first, then heteroskedasticity:

```
two_step_sequential(@formula(y ~ x1 + x2), data_hae; Z = @formula(0 ~ abs(x1) + x2))
```

```
Coefficients:
```

	Coef.	Std. Error	t	Pr(>|t|)	Lower 95%	Upper 95%
(Intercept)	1.0166	0.331721	3.06	0.0023	0.364854	1.66835
x1	1.9606	0.0757041	25.90	<1e-93	1.81186	2.10933
x2	-0.955632	0.0655662	-14.58	<1e-39	-1.08445	-0.826811

```
AR(1) coefficient:
```

	Coef.	Std. Error	z	Pr(>|z|)	Lower 95%	Upper 95%
ρ	0.745123	0.0298259	24.98	<1e-99	0.686666	0.803581

```
Log-variance equation (gamma):
```

	Coef.	Std. Error	z	Pr(>|z|)	Lower 95%	Upper 95%
log σ_0^2	0.0337159	0.173542	0.19	0.8460	-0.30642	0.373852
z1	0.857706	0.160573	5.34	<1e-07	0.542989	1.17242
z2	0.143196	0.10375	1.38	0.1675	-0.0601497	0.346542

```
N: 500   R^2: 0.5228   Adj. R^2: 0.5209
```

Joint HARE – estimates AR(1) and heteroskedasticity simultaneously:

```
two_step_joint(@formula(y ~ x1 + x2), data_hae; Z = @formula(0 ~ abs(x1) + x2))
```

```
Coefficients:
```

	Coef.	Std. Error	t	Pr(>|t|)	Lower 95%	Upper 95%
(Intercept)	1.04166	0.234634	4.44	<1e-04	0.580665	1.50266
x1	1.96357	0.055024	35.69	<1e-99	1.85547	2.07168
x2	-0.964916	0.049563	-19.47	<1e-62	-1.06229	-0.867537

```
AR(1) coefficient:
```

	Coef.	Std. Error	z	Pr(>|z|)	Lower 95%	Upper 95%
ρ	0.73498	0.030325	24.24	<1e-99	0.675544	0.794416

```
Log-variance equation (gamma):
```

	Coef.	Std. Error	z	Pr(>|z|)	Lower 95%	Upper 95%
log σ_0^2	0.1395	0.108134	1.29	0.1970	-0.0724395	0.351439
z1	0.712469	0.101468	7.02	<1e-11	0.513596	0.911342
z2	0.00611855	0.0678506	0.09	0.9281	-0.126866	0.139103

```
Log-likelihood: -439.3689
N: 500   R^2: 0.5234   Adj. R^2: 0.5214
```

Heteroskedasticity Tests

HARE provides two score (LM) tests for heteroskedasticity that are unique to the package. Both test $H_0 : \gamma = 0$ (homoskedasticity) and are distributed as $\chi^2(p)$ under H_0, where p is the number of variance regressors (columns of Z excluding the intercept).

We reuse the exponential-variance dataset from the first section ($n = 500$, $\log \sigma_i^2 = x_{1i}$), so both tests should reject strongly. The R equivalents are `skedastic::harvey()` and `skedastic::glejser()`.

Harvey test – LM statistic ESS/trigamma(1/2) from the auxiliary regression of $\log \hat{u}_i^2$ on Z:

```
y_test   = b[1] .+ b[2] .* x1 .+ b[3] .* x2 .+ sigma_exp .* randn(nobs);
data_test = (y = y_test, x1 = x1, x2 = x2);

harvey_test(@formula(y ~ x1 + x2), data_test)
```

```
Harvey test for multiplicative heteroskedasticity (exponential link)
  H₀: γ = 0  in  log(σᵢ²) = γ₀ + zᵢ'γ
  LM = 0.4511   df = 2   p-value = 0.7981
```

Glejser test – LM statistic from the auxiliary regression of $|\hat{u}_i|$ on Z:

```
glejser_test(@formula(y ~ x1 + x2), data_test)
```

```
Glejser test for heteroskedasticity (quadratic link / linear SD)
  H₀: γ = 0  in  σᵢ = γ₀ + zᵢ'γ
  LM = 2.3974   df = 2   p-value = 0.3016
```

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